Geometric Twins – Homework Answer Key

CBSE Class 7 Mathematics · Ganita Prakash (Part 2) · Chapter: Geometric Twins

Section A – Congruent Figures

Q1. shape; size.

Q2. (a) True (b) False – squares of different side lengths are not congruent. (c) True (d) False – the size changes.

Q3. Pairs **(a)** and **(c)** are congruent. (a) Both rectangles are 3 × 2 (one is turned). (c) Both right triangles have sides 3 and 2 units along the grid (one is flipped). (b) Squares of side 2 and 3 – different sizes. (d) 4 × 1 rectangle and 2 × 2 square – different shapes.

Q4. (a) The radius (or diameter). (b) The length of one side. (c) The length and the breadth.

Q5. (a) Yes – angles with equal measures are congruent. (b) No – their lengths are different.

Q6. Answers may vary. For example: two ₹10 coins of the same design, two sheets of A4 paper, two floor tiles of the same design, two pages of the same notebook.

Section B – Corresponding Parts

Q7. (a) PQ (b) QR (c) ∠Q (d) ∠R

Q8. LM = DE = 6 cm, MN = EF = 7 cm, LN = DF = 5 cm.

Q9. ∠K = ∠X = 60°, ∠L = ∠Y = 70°, ∠M = ∠Z = 180° − (60° + 70°) = 50°.

Q10. ΔABC ≅ ΔMNL. (AB = MN = 4 cm, BC = NL = 5 cm, CA = LM = 6 cm, so A ↔ M, B ↔ N, C ↔ L.)

Q11. (b) ∠A = ∠P.

Section C – Conditions for Congruence of Triangles

Q12. (a) Side–Side–Side (b) Side–Angle–Side (c) Angle–Side–Angle (d) Right angle–Hypotenuse–Side

Q13. Yes. AB = DE = 5 cm, BC = EF = 6 cm, AC = DF = 4.5 cm. Condition: **SSS**. ΔABC ≅ ΔDEF.

Q14. Yes. PQ = XY = 4 cm, ∠Q = ∠Y = 60°, QR = YZ = 5 cm. The 60° angle is between the two sides. Condition: **SAS**. ΔPQR ≅ ΔXYZ.

Q15. Yes. Condition: **ASA** (the 6 cm side lies between the 50° and 70° angles). ΔABC ≅ ΔDEF.

Q16. Yes. Both have a right angle, equal hypotenuses (AC = PR = 10 cm) and one more equal side (AB = PQ = 6 cm). Condition: **RHS**. ΔABC ≅ ΔPQR.

Q17. included between (the angle formed by the two sides).

Q18. (a) SSS (b) SAS (c) ASA (d) RHS

Q19. No. They have the same shape but not the same size (ΔPQR is bigger). Equal angles alone (AAA) do not make triangles congruent.

Q20. (a) **SSS** – AB = AD, CB = CD and AC is common to both triangles. (b) ∠DAC = ∠BAC = 35° (corresponding parts of congruent triangles).

Q21. (a) They are vertically opposite angles. (b) **SAS** – OA = OB, ∠AOC = ∠BOD, OC = OD. (c) BD = AC = 5 cm.

Q22. Yes. AB = CD and BC = DA (sides of a square), and AC is common. Condition: **SSS**. (SAS can also be used: AB = CD, ∠B = ∠D = 90°, BC = DA.)

Section D – Isosceles and Equilateral Triangles

Q23. ∠C = ∠B = 65° (angles opposite equal sides are equal). ∠A = 180° − (65° + 65°) = 50°.

Q24. 60° (180° ÷ 3).

Q25. (a) PQ = PR (sides opposite the equal angles). (b) ∠P = 180° − (70° + 70°) = 40°.

Q26. The other two angles are equal: (180° − 80°) ÷ 2 = 50° each.

Q27. Third angle = 180° − (45° + 45°) = 90°. It is a **right-angled isosceles triangle**.

Q28. No. 95° + 95° = 190°, which is more than 180°. The sum of the angles of a triangle is always 180°.

Section E – Congruence Around Us

Q29. (a) **RHS** – ∠PQA = ∠PQB = 90°, hypotenuse PA = PB, and PQ is common. (b) QB = QA = 3 m.

Q30. Yes. All three sides of one triangle are equal to the three sides of the other (3 cm, 4 cm, 5 cm), so they are congruent by **SSS**. The order in which the sides are listed does not matter.